

1 (a) It is given that $\frac{2px+9qy}{2py+qx} = 3$, p and q are constants and $2p \neq 3q$.

(i) Show that $x = 3y$. [2]

(ii) Evaluate $\frac{x+y}{y}$.

Answer _____ [2]

(b) Simplify $\frac{3a + 7b}{16a^2 - 49(a + b)^2}$.

Answer _____ [3]

(c) Solve the equation $2^{x+3} = 320 - 2^{x+1}$.

Answer $x =$ _____ [3]

- 2 The Singapore Indoor Stadium has a seating capacity of 12 000. For shows in February 2020, tickets for Disney on Ice are priced at \$50 per adult and \$35 per child.

(a) For one of the evening shows in February, 85% of the seats in the stadium are occupied. 3200 of the people present are children. Calculate the total amount of money collected from the sale of tickets for that evening.

Answer \$ _____ [2]

(b) In light of the Covid-19 situation, the price of the adult ticket for a show in February 2020 is 20% cheaper than a show in December 2019. Calculate the price of the adult ticket in December 2019.

Answer \$ _____ [2]

(c) The amount of money collected from the sales of tickets from one day in February 2020 was \$375 000. This sum of money is divided among cost of operation, wages and profit in the ratio 3 : 5 : 7.
The profit is invested at a rate of 4% per annum compounded quarterly for a period of 2 years.
Calculate the compound interest earned.

Answer \$ _____ [3]